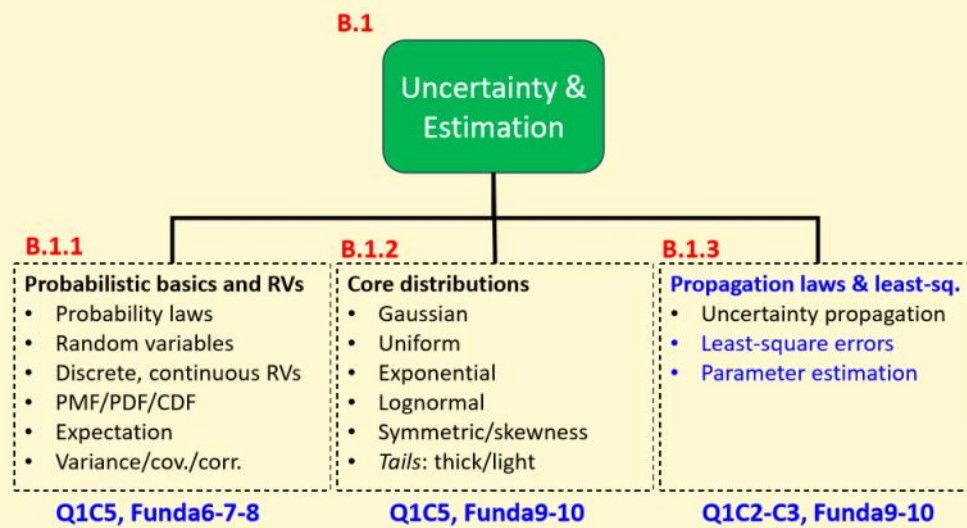


L10

Sep 22, 2026

Module Overview



Linear models: intro

- Linear functional relationship

$$E[Y|X=x] = Ax$$

deterministic constant

$$Y = Ax + \varepsilon$$

such that, $E[\varepsilon] = 0$

$$\sigma_Y = \sigma_\varepsilon \quad (\text{As } A \text{ is a constant \& } x \text{ is given})$$

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Linear models: example

- Linear function relationship

$$E[Y_i] = \beta_0 + \beta_1 x_i$$

t_i : observation time ; independent variable

x_1 : Intercept

x_2 : Slope (rate of change)

Linear functional model is

$$E \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_{10} \end{bmatrix} = \underbrace{\begin{Bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{10} \end{Bmatrix}}_Y = \underbrace{\begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_{10} \end{bmatrix}}_A \underbrace{\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}}_{x \quad 2 \times 1}$$

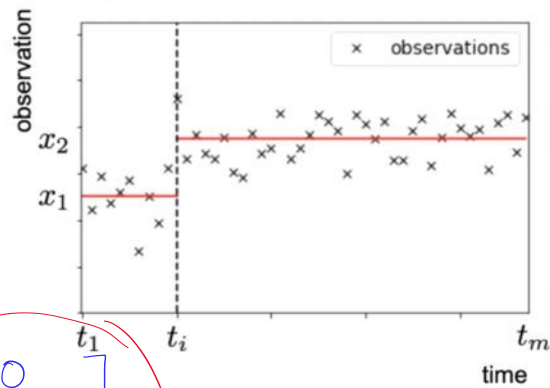
$Y \quad A \quad x$

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Linear models: example

- Is this a linear model?

Write its linear functional relationship



$$E \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_i \\ Y_{i+1} \\ \vdots \\ Y_m \end{pmatrix}_{m \times 1} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{bmatrix}_{m \times 2} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}_{2 \times 1}$$

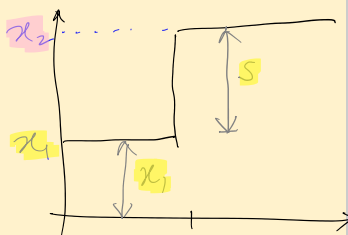
A

Linear models: example

- Is this a linear model

Step function, $s = x_2 - x_1$

Write its linear functional relationship.



$$E \begin{pmatrix} Y_1 \\ \vdots \\ Y_i \\ Y_{i+1} \\ \vdots \\ Y_n \end{pmatrix}_{n \times 1} = \begin{bmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{bmatrix}_{n \times 2} \begin{Bmatrix} x_1 \\ s \end{Bmatrix}_{2 \times 1}$$

$Y = A x$

Linear models: example

- Is this a linear model

$$E[Y_i] = x_1 + x_2 t_i + x_3 t_i^2$$

Write its linear functional relationship

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Linear models: example

- Is this a linear model

$$E[Y_i] = \beta_0 \cdot x_{1,i}^{\beta_1} \cdot x_{2,i}^{\beta_2}$$

Write its linear functional relationship

$$E \begin{bmatrix} \ln Y_1 \\ \vdots \\ \ln Y_n \end{bmatrix} = \begin{bmatrix} 1 & \ln x_{1,1} & \ln x_{2,1} \\ \vdots & \vdots & \vdots \\ 1 & \ln x_{1,n} & \ln x_{2,n} \end{bmatrix} \begin{bmatrix} \ln \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

A
 X

Y
 X

x_i 's are coefficients (params) yet to be determined

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Elements of models ^{Modeling}

0. Eyeball the Data

Scatter plot, histogram, change the scale $\leftarrow \begin{matrix} \log-\log \\ \text{semi-log } X \\ \text{semi-log } Y \end{matrix}$

1. Estimation Model

$$Y = A X + \epsilon \quad \Bigg| \quad Y = X \beta + \epsilon$$

2. Inference Predicted

Y dependent variable

3. Prediction

Predictor

A X ind./explanatory/covariate

Parameters

α β

4. Explanation

Estimating Parameters 'aka'

Model Building Phase'

5. Diagnosis

Role of A X (independent variable)
is NOT to predict Y , yet.

Instead, it is to estimate parameters.

Typical Steps in Practice

1. Plan the experiment. Decide $\left. \begin{matrix} \text{input variables} \\ \text{output variables} \\ \text{number of tests} \end{matrix} \right\} \rightarrow (t_i, y_i)_{i \in \{1, 2, \dots, n\}}$
2. Hypothesize a Model
 $Y = \beta_0 + \beta_1 t + \beta_2 t^2$ (...say)
3. Estimate the parameters, i.e.; $\beta_0, \beta_1, \beta_2$
4. Use the model to predict Y for different values of t .

Elements of models

Linear Regression with R
Julian J. Faraway

0. Eyeball the data.

Scatter plot, histogram, change scale

1. Estimation GOAL: obtain parameter estimates ($\hat{\beta}$)

concept: least squares, method of moments, maximum likelihood estimation

2. Inference GOAL: compare competing model; uncertainty in parameter

concept: confidence interval, hypothesis testing, p-values $\hat{\beta}$ (CI) (HT)

3. Prediction GOAL: forecast new outcomes

concept: Mean square error, CI for $\hat{y}$

4. Explanation GOAL: Interpret the fitted model

concept: Feature dependence, causality

5. Diagnosis GOAL: Assess model assumptions

concept: constant variance, outliers treatment

Least-square errors

$$y = \beta_0 x_1 + \beta_2 x_2 \rightarrow \text{Non-linear least-square}$$

A linear model with a single feature has two parameters:

Intercept slope

An example:

$$y_1 = m x_1 + c$$

$$y_2 = m x_2 + c$$

...

$$y_5 = m x_5 + c$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_5 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_5 \end{bmatrix} \times \begin{bmatrix} m \\ c \end{bmatrix}$$

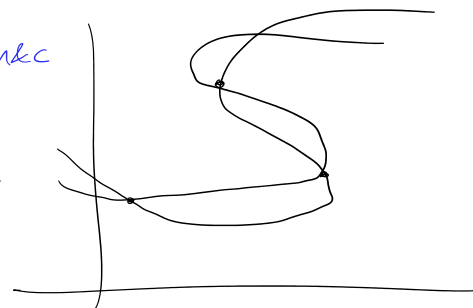
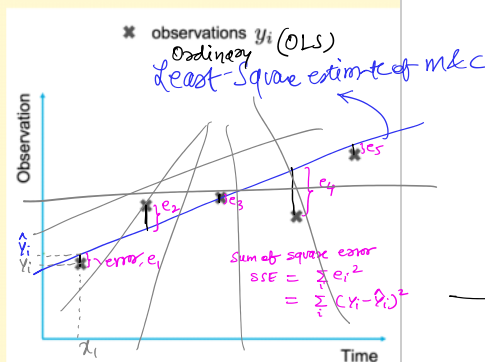
$$\Rightarrow y_{5 \times 1} = A_{5 \times 2} x_{2 \times 1} \rightarrow \text{No solution}$$

$$\Rightarrow (A^T)_{2 \times 5} y_{5 \times 1} = (A^T)_{2 \times 5} A_{5 \times 2} x_{2 \times 1} \rightarrow \text{Left Multiply with } A^T$$

$$\Rightarrow (A^T y)_{2 \times 1} = (A^T A)_{2 \times 2} x_{2 \times 1}$$

$$\Rightarrow (A^T A)^{-1} (A^T y) = x_{2 \times 1} \rightarrow \text{Left Multiply with } (A^T A)^{-1}, \text{ if it exists}$$

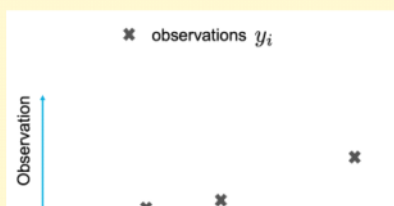
$$\Rightarrow x_{2 \times 1} = (A^T A)^{-1} A^T y$$



Least-square errors

A linear model with a single feature has two parameters:

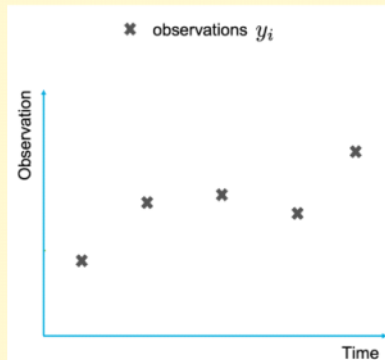
An example:



Least-square errors

A linear model with a single feature has two parameters:

An example:



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Least-square errors

$$E[Y | X = x] = Ax$$

Observed

(x_i, y_i)

predicted

$\hat{y}_i = A \hat{x}_i$

$$\hat{\epsilon}_i = y_i - \hat{y}_i = y_i - A \hat{x}_i$$

2- norm error

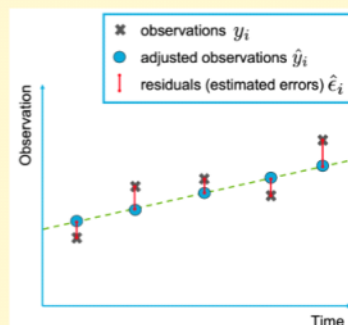
$$\| \epsilon \| = \sqrt{\epsilon_1^2 + \epsilon_2^2 + \dots + \epsilon_n^2}$$

$$y = 3x$$

$y=6$ $x=2$

$$y = Ax + \epsilon$$

$\| \epsilon \|_1 = |\epsilon_1| + |\epsilon_2| + \dots + |\epsilon_n|$
 1- norm: Manhattan Norm
 2- norm: Euclidean Norm



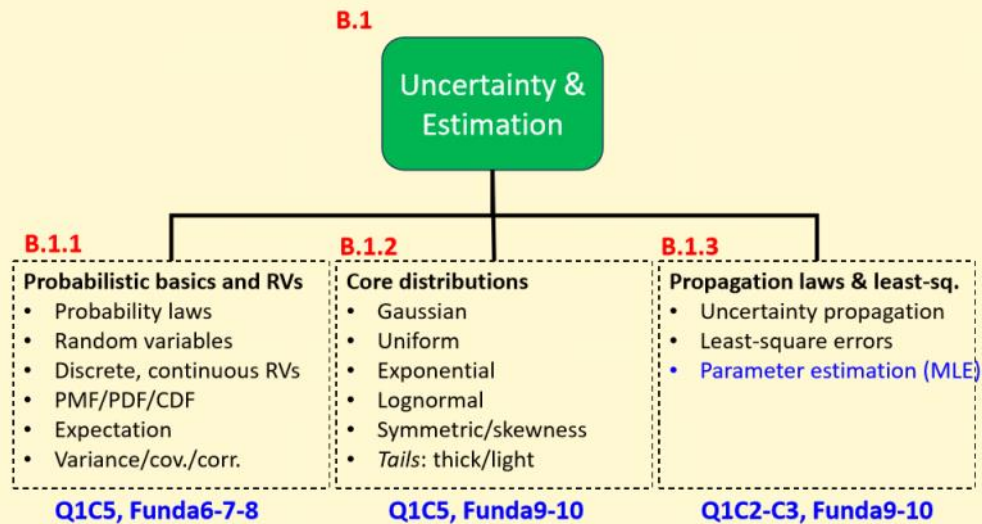
Parameter estimation

- Upcoming topics:
 - **Estimation**: Maximum likelihood estimation
 - **Inference**: Confidence interval
 - **Inference**: Hypothesis testing
 - **Goodness of fit**: χ^2 , KS

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L11
Sep 23, 2026

Module Overview



Flow (so far and week ahead)

- So far:
 - Probability axioms, rules, PMF/PDF/CDF
 - Variance, covariance, **correlation**
 - **Distributions**: uniform, Gaussian, exponential, Gumbel, lognormal
 - Standard normal; reading **z-table**
 - **Uncertainty propagation**: linear, nonlinear function (FO and SO approximation)
 - Fitting a distribution **Linear regression** (estimation): **least-square method**
- This week:
 - **Estimation**: Method of moments (**MoM**), Maximum likelihood estimation (**MLE**)
 - **Inference**: Confidence interval (**CI**)
 - **Inference**: Hypothesis testing (**HT**)
 - **Goodness of fit (GoF)**: χ^2 , KS
- Check-in with textbook
 - MLE: Q1C3.6 and Q1C5.4.2
 - CI: Q1C3.5
 - HT: Q1C3.9
 - GoF: Q1C5.4

OLS ← m, c

MoM

MLE

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Parameter estimation: method of moments

- Obtain the moments (e.g., mean, variance) from two sources:

- **Observations:**

$$E[X] = \mu_x = \int_{-\infty}^{\infty} x f_x(x) dx$$

$$\text{Var}[X] = \sigma_x^2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 f_x(x) dx$$

$$\text{Var}[X] = \sigma_x^2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 f_x(x) dx$$

• Parameters:

$$\theta[X] = \frac{1}{\sigma^3} \int_{-\infty}^{\infty} (x - \mu_x)^3 f_x(x) dx$$

• Equate them to solve for the parameters

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Distributions and parameters

Dist.	PDF	CDF	Mean & Variance
Normal/ Gaussian	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$	$F(x) = \Phi(x)$ Use z-table	$E[X] = \mu$ $\text{Var}[X] = \sigma^2$
Uniform	$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$	$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{otherwise} \end{cases}$	$E[X] = \frac{1}{2}(a+b)$ $\text{Var}[X] = \frac{1}{12}(b-a)^2$
Exponential	$f(x) = \lambda \exp(-\lambda x)$	$F(x) = 1 - \exp(-\lambda x)$	$E[X] = \frac{1}{\lambda}$ $\text{Var}[X] = \frac{1}{\lambda^2}$
Gumbel	$f(x) = \frac{1}{\beta} \exp[-z + \exp(-z)]$, where $z = \frac{x-\alpha}{\beta}$	$F(x) = \exp[-\exp(-z)]$	$E[X] = \alpha + \beta\gamma$ $\text{Var}[X] = \frac{\pi^2}{6}\beta^2$ $\gamma = 0.577$
Lognormal	$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2\right]$	$F(x) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$ Use z-table	$E[X] = \exp\left(\mu + \frac{\sigma^2}{2}\right)$ $\text{Var}[X] = [\exp(\sigma^2) - 1] \exp(2\mu + \sigma^2)$

Example

- Assume that Earthquakes in Rome expressed as MCS index follow Gumbel type-I distribution. Earthquake intensity and number of earthquakes between 1000 CE – 1980 CE are:

MCS intensity, x_i	2	3	4	5	6	7	$\sum x = \sum x_i f_i$	$\sum x^2 = \sum x_i^2 f_i$	$\sum f_i$
Number, f_i	113	132	56	22	4	2	994	3328	329

Find the parameters of the distribution.

- Observations:

$$E[X] = \sum x / \sum f_i = 994 / 329 \approx 3.02$$

$$\text{Var}[X] = E[X^2] - (E[X])^2 \approx \frac{3328}{329} - 3.02^2 \approx 0.995$$

- Parameters for Gumbel type-I:

$$\text{Var}[X] = \frac{\pi^2}{6} \beta^2 \approx 0.995 \quad \gamma = 0.577$$

$$E[X] = \alpha + \beta \gamma \approx 3.02$$

- Equate them to solve for the parameters

$$\Rightarrow \beta \approx 0.77 \quad \Rightarrow \alpha \approx 2.57$$

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Objective of Parameter Estimation:

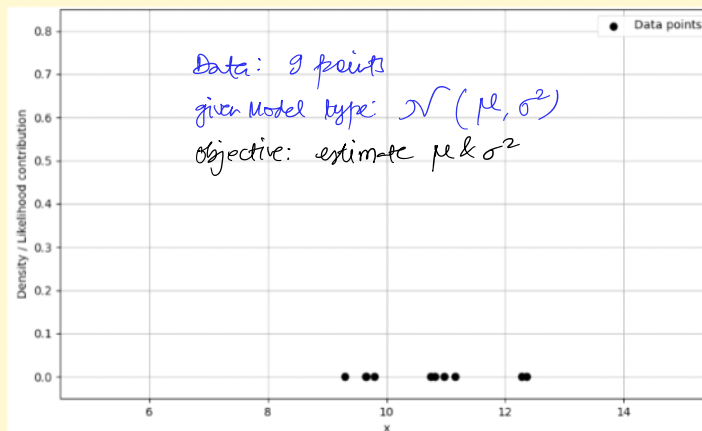
Given $\left\{ \begin{array}{l} 1. \text{ Data} \\ k \\ 2. \text{ Model} \end{array} \right\}$, Estimating "Model Parameters." $\left\{ \begin{array}{l} \text{OLS} \\ \text{MoM} \\ \text{MLE} \end{array} \right\}$

Maximum likelihood estimation

Basic idea

Objective of MLE:

To estimate model parameters such that they maximize the "likelihood" of observed data.



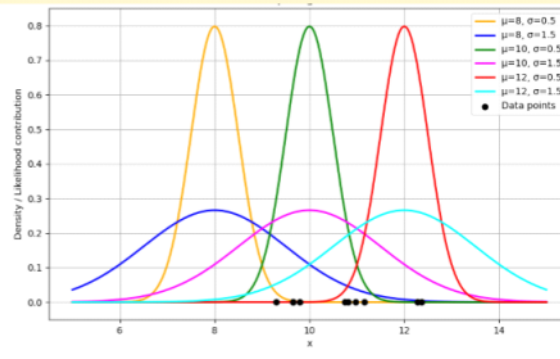
Maximum likelihood estimation

$$\mathcal{L}(\hat{\mu}, \hat{\sigma} | x_i) = f(x_i | \hat{\mu}, \hat{\sigma})$$

Maximize this likelihood function

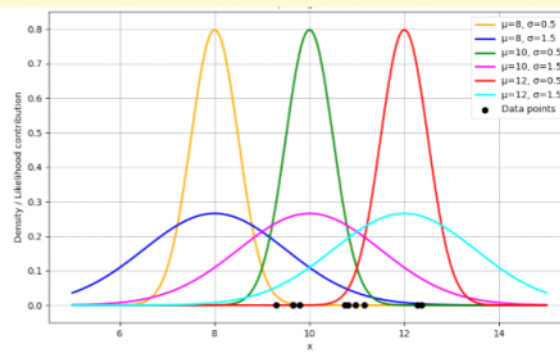
$$\mathcal{L}(\hat{\mu}, \hat{\sigma} | \underline{x}) = \prod_{i=1}^n f(x_i | \hat{\mu}, \hat{\sigma})$$

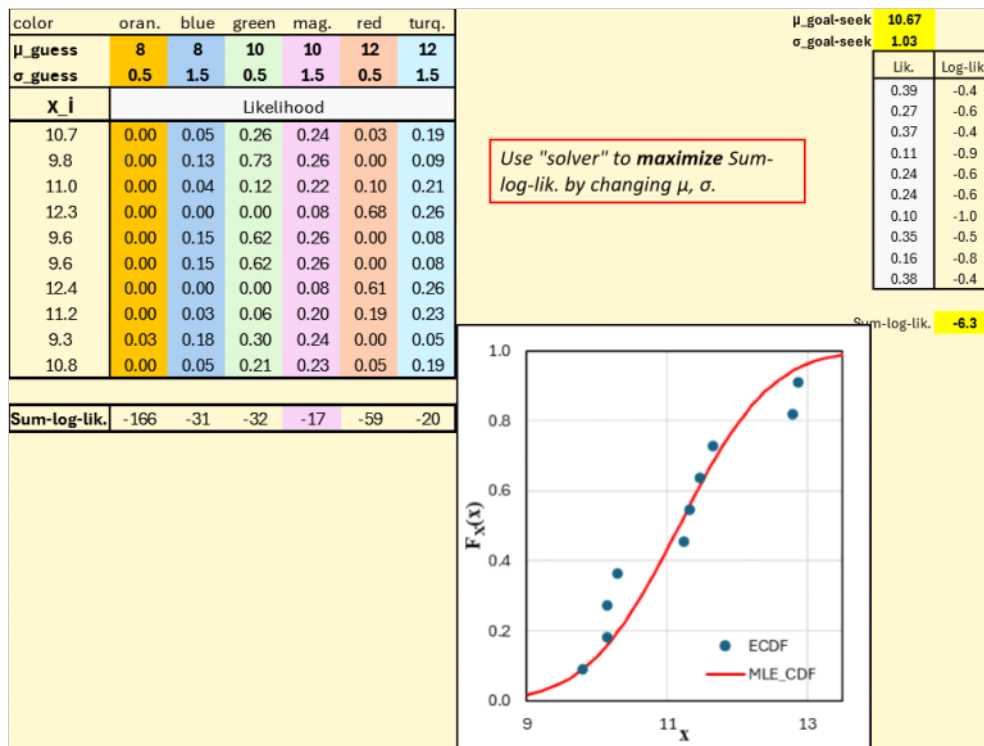
$$\hat{\mu}, \hat{\sigma} = \operatorname{argmax} \mathcal{L}(\hat{\mu}, \hat{\sigma} | \underline{x})$$



Maximum likelihood estimation

Excel Demo!





Questions, comments,
or concerns?